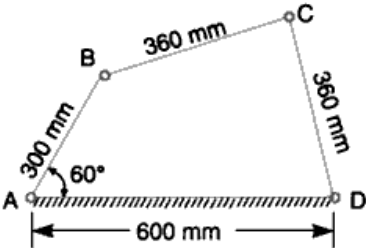
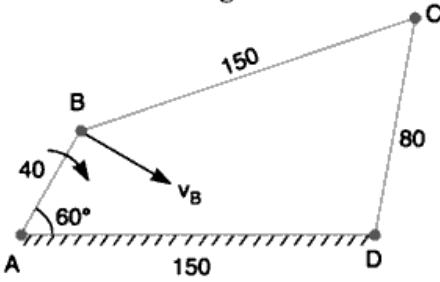
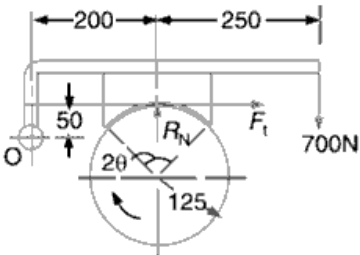


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B.Tech. Branch : Mech. Engg./Mech. Engg. Sandwich Sem :IV Subject Code & Name: BTMC402 - Theory of Machines-I Max Marks: 60 Date: 18 /01/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve any 2 from the following questions.		12
A)	Discuss double slider crank chain mechanism & its inversion	CO1	6
B)	Fig. shows $AB = 300$ mm, $BC = CD = 360$ mm, and $AD = 600$ mm. The angle $BAD = 60^\circ$. The crank AB rotates uniformly at 100 r.p.m. Locate all the instantaneous centres and find the angular velocity of the link BC 	CO1	6
C)	In a four bar chain $ABCD$, AD is fixed and is 150 mm long. The crank AB is 40 mm long and rotates at 120 r.p.m. clockwise, while the link $CD = 80$ mm oscillates about D. BC and AD are of equal length. Find the angular velocity of link CD when angle $BAD = 60^\circ$. 	CO1	6
Q.2	Solve any 2 from the following questions.		12
A)	Describe uniform pressure theory for flat pivot bearing	CO2	6
B)	A conical pivot supports a load of 20 kN, the cone angle is 120° and the intensity of normal pressure is not to exceed 0.3 N/mm ² . The external diameter is twice the internal diameter. Find the outer and inner radii of the bearing surface. If the shaft rotates at 200 r.p.m. and the coefficient of friction is 0.1, find the power absorbed in friction. Assume uniform pressure.	CO2	6
C)	Explain in brief Hydrostatic and hydrodynamic lubrication.	CO2	6
Q. 3	Solve any 2 from the following questions.		12
A)	Explain in brief different types of absorption type dynamometers	CO3	6

B)	A single block brake is shown in Fig. The diameter of the drum is 250 mm and the angle of contact is 90°. If the operating force of 700 N is applied at the end of a lever and the coefficient of friction between the drum and the lining is 0.35, determine the torque that may be transmitted by the block brake. 	CO3	6
C)	Explain the construction & working of centrifugal Clutch	CO3	6
Q.4	Solve any 2 from the following questions.		12
A)	Explain with figures any three types of followers.	CO4	6
B)	A cam is to give the following motion to a knife-edged follower : 1. Outstroke during 60° of cam rotation; 2. Dwell for the next 30° of cam rotation; 3. Return stroke during next 60° of cam rotation, and 4. Dwell for the remaining 210° of cam rotation. The stroke of the follower is 40 mm and the minimum radius of the cam is 50 mm. The follower moves with uniform velocity during both the outstroke and return strokes. Draw the profile of the cam when (a) the axis of the follower passes through the axis of the cam shaft, and (b) the axis of the follower is offset by 20 mm from the axis of the cam shaft.	CO4	6
C)	Explain any three terms used in cam along with sketch.	CO4	6
Q. 5	Solve any 2 from the following questions.		12
A)	Explain the method of balancing of several masses in the same planes	CO5	6
B)	A shaft carries four masses A, B, C and D of magnitude 200 kg, 300 kg, 400 kg and 200 kg respectively and revolving at radii 80 mm, 70 mm, 60 mm and 80 mm in planes measured from A at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are A to B 45°, B to C 70° and C to D 120°. The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100 mm, between X and Y is 400 mm and between Y and D is 200 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions.	CO5	6
C)	Explain the direct and reverse crank method for determining unbalanced forces in radial engines.	CO5	6
	*** End ***		